AS49-A017

Impact of Atmosphere-Ocean Coupling on Regional Climate Downscaling over East Asia

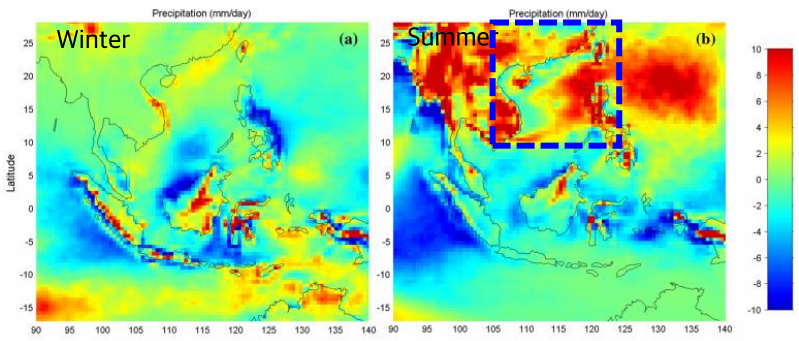
Junseo PARK+, Woojin CHO, Seok-Woo SHIN, Taehyung KIM, Dong-Hyun CHA#

Ulsan National Institute of Science and Technology (UNIST)

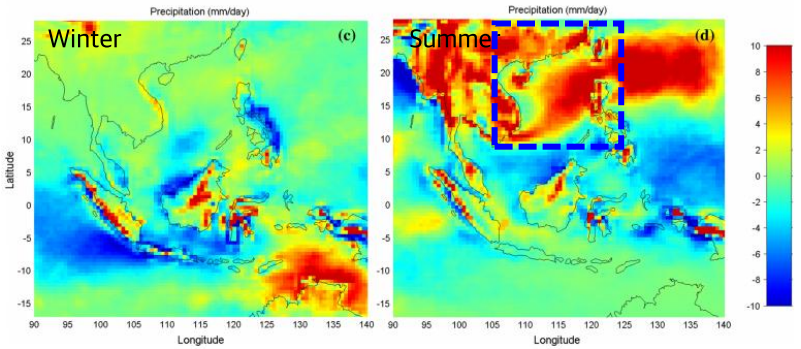
1. Introduction

1-1. Motivation: Why Coupling?

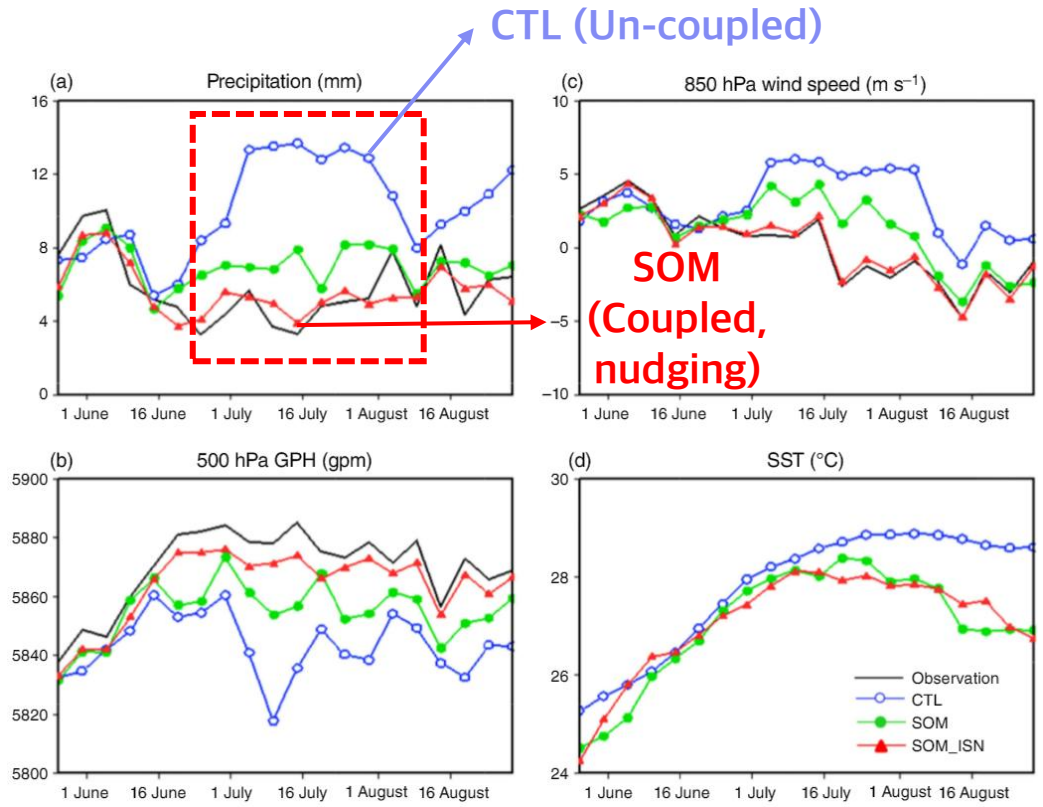
Coupled RCM (RegCM3 & SOM)



RegCM3-only



Bias of Coupled Regional Climate Model (RCM) and uncoupled RCM (Wei et al., 2013)

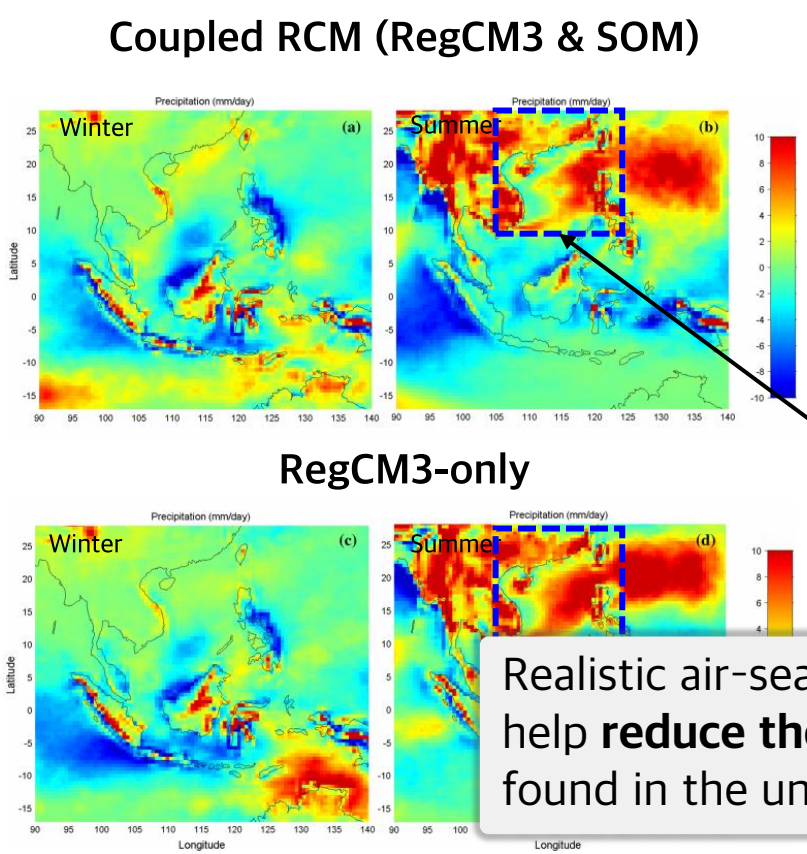


Time series of simulations from the Coupled and the uncoupled RCM (Cha et al., 2015)

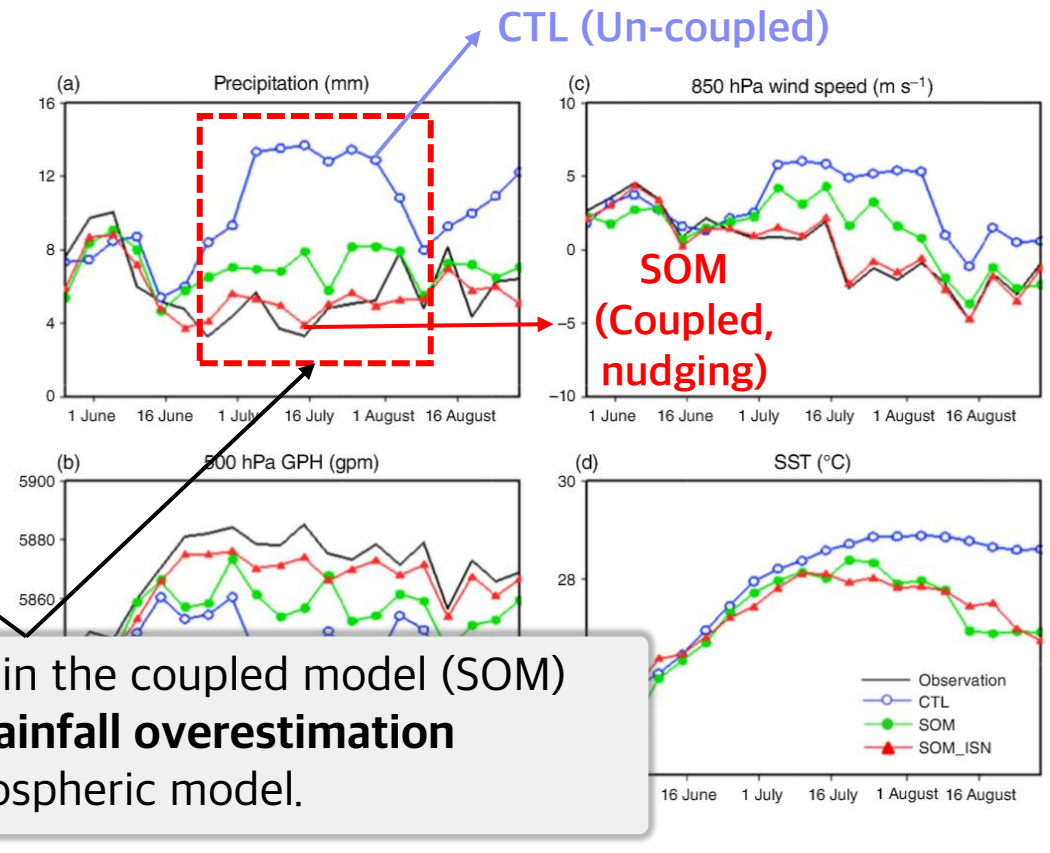
Wei, J., Malanotte-Rizzoli, P., Eltahir, E. A., Xue, P., & Xu, D. (2014).
Coupling of a regional atmospheric model (RegCM3) and a regional oceanic model (FVCOM) over the maritime continent.
Climate Dynamics, 43(5), 1575-1594.
Cha, D. H., Jin, C. S., Moon, J. H., & Lee, D. K. (2016).
Improvement of regional climate simulation of East Asian summer monsoon by coupled air--sea interaction and large-scale nudging.
International Journal of Climatology, 36(1).

RegCM3: Regional Climate Model version 3
SOM: Slab Ocean Model
RCM: Regional Climate Model

1-1. Motivation: Why Coupling?



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Time series of simulations from the Coupled and the uncoupled RCM
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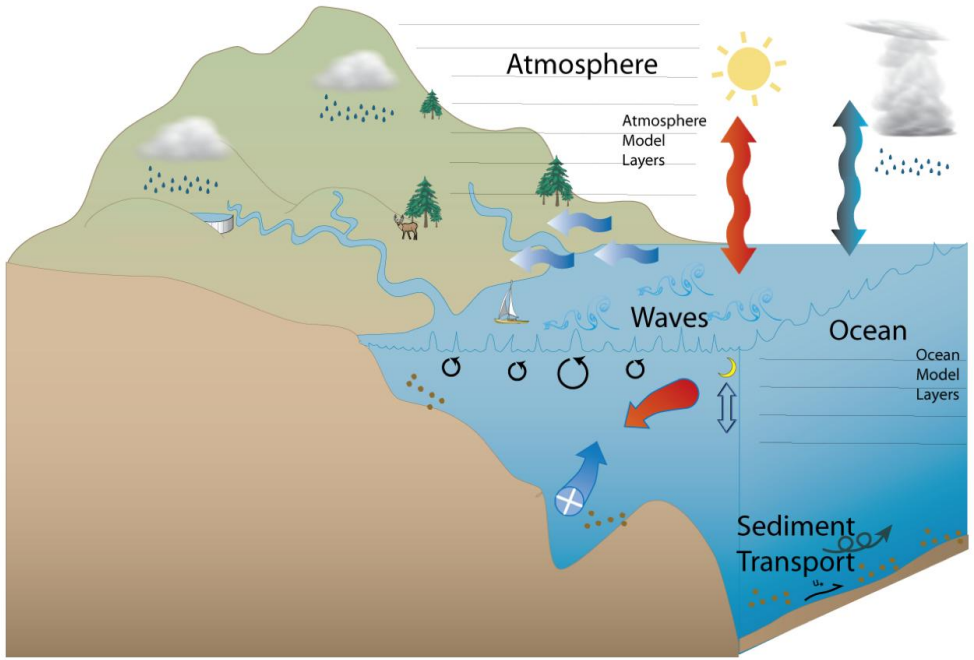
Realistic air-sea interactions in the coupled model (SOM) help **reduce the monsoon rainfall overestimation** found in the uncoupled atmospheric model.

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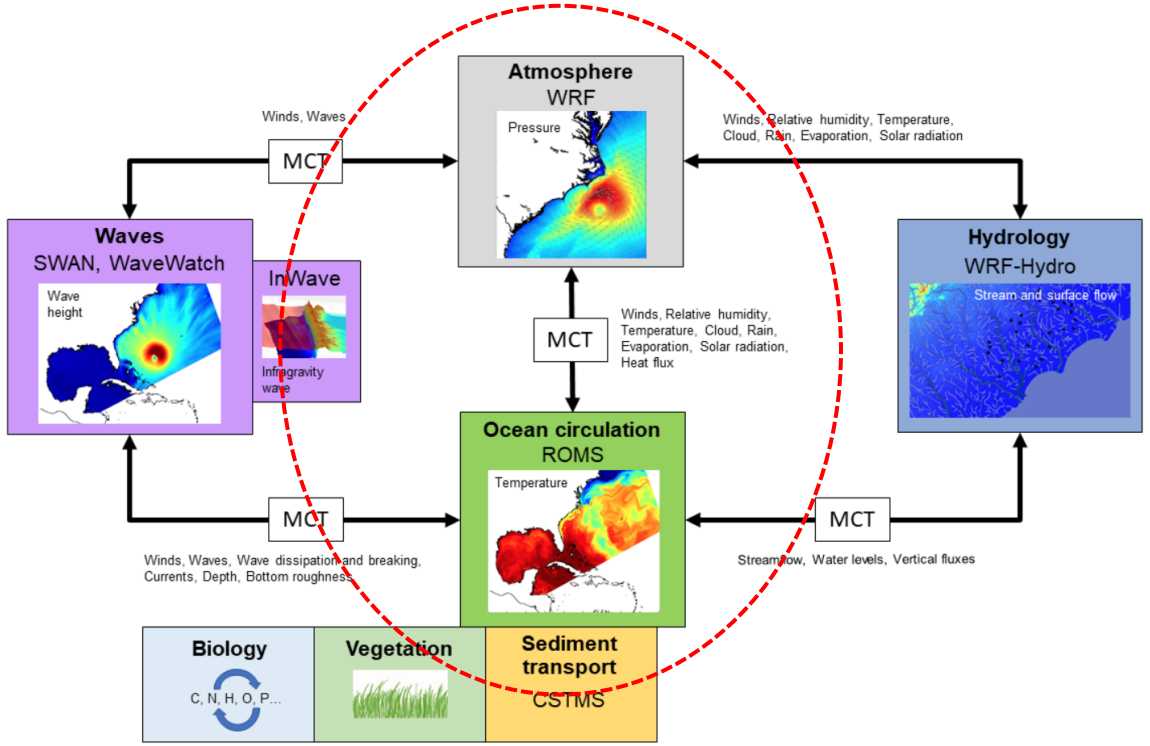
RegCM3: Regional Climate Model version 3
SOM: Slab Ocean Model
RCM: Regional Climate Model

1-2. Chosen Model

Coupled Ocean Atmosphere Wave Sediment Transport (COAWST)



COAWST Modeling System infographic
By Coastal and Marine Hazards and Resources Program (2015)



COAWST Modeling System
By Woods Hole Coastal and Marine Science Center

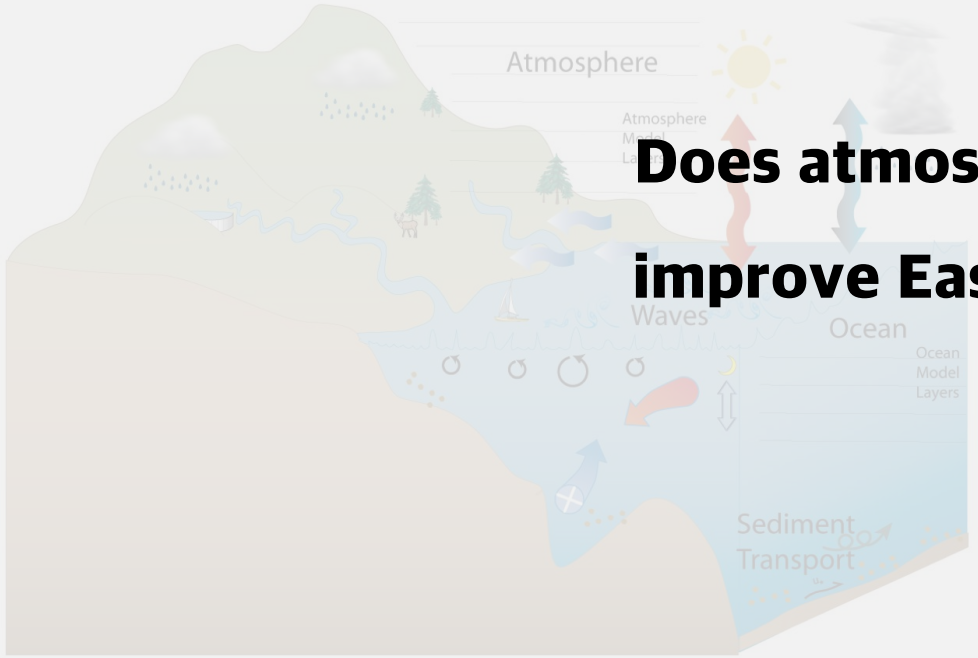
WRF: Weather Research and Forecast
ROMS: Regional Ocean Modeling System
MCT: Model Coupling Toolkit

1-2. Chosen Model

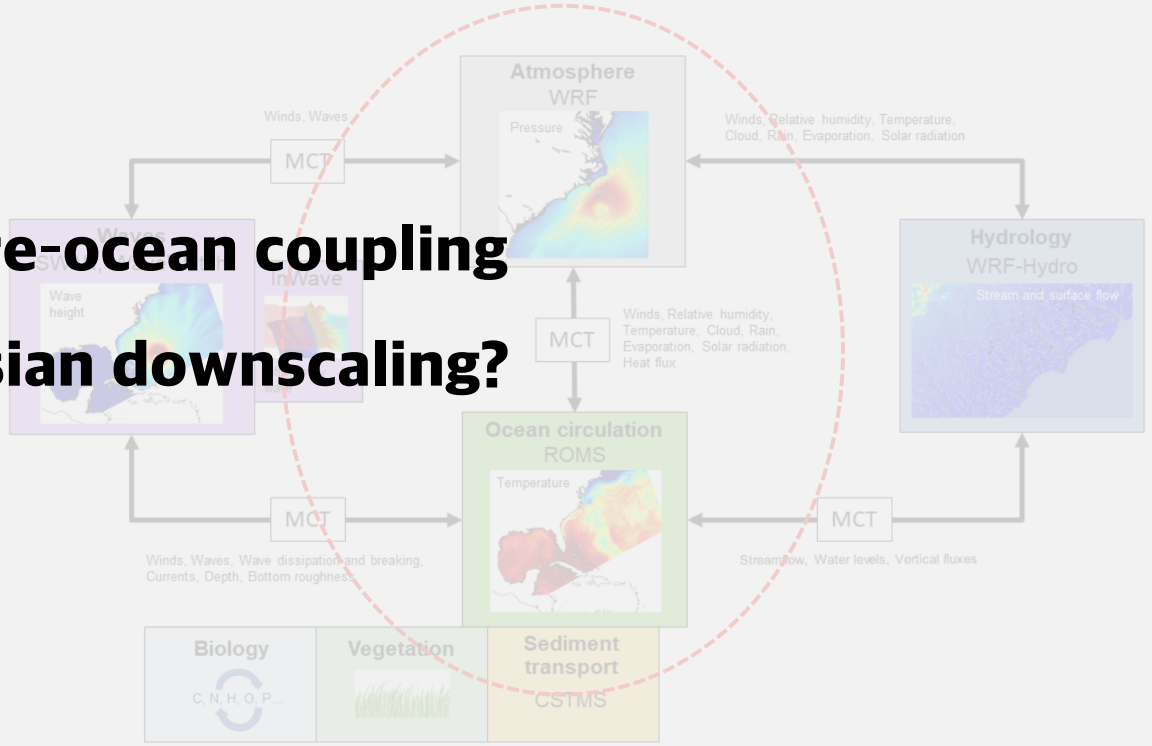
Coupled Ocean Atmosphere Wave Sediment Transport (COAWST)

Does atmosphere-ocean coupling

improve East-Asian downscaling?



COAWST Modeling System infographic
By Coastal and Marine Hazards and Resources Program (2015)

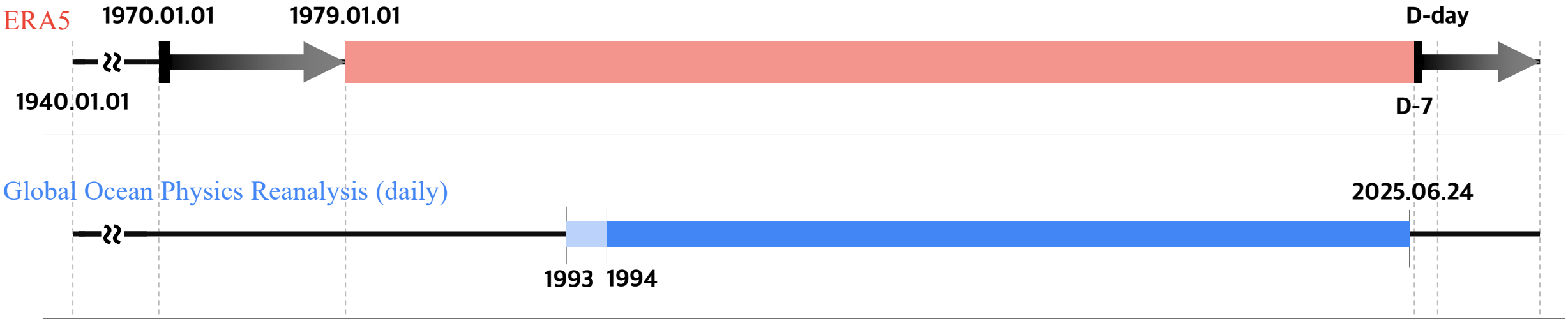


COAWST Modeling System
By Woods Hole Coastal and Marine Science Center

WRF: Weather Research and Forecast
ROMS: Regional Ocean Modeling System
MCT: Model Coupling Toolkit

2. Data & Method

2-1. Boundary Data



GLORYS Data Description

Spatial extent: Global Ocean (Lat -80° to 90°, Lon -180° to 179.92°)

Spatial resolution: 0.083° × 0.083° (1/12°, horizontal)

Temporal extent: 1 Jan 1993 to 24 Jun 2025

Temporal resolution: Daily / Monthly

Elevation (depth) levels: 50

ECMWF: European Centre for Medium-Range Weather Forecasts
ERA5: ECMWF Reanalysis version 5
GLORYS: Global Ocean Physics Reanalysis

2-1. Precipitation Data

MSWEP (Multi-Source Weighted-Ensemble Precipitation)



MSWEP Description

Spatial extent: Global

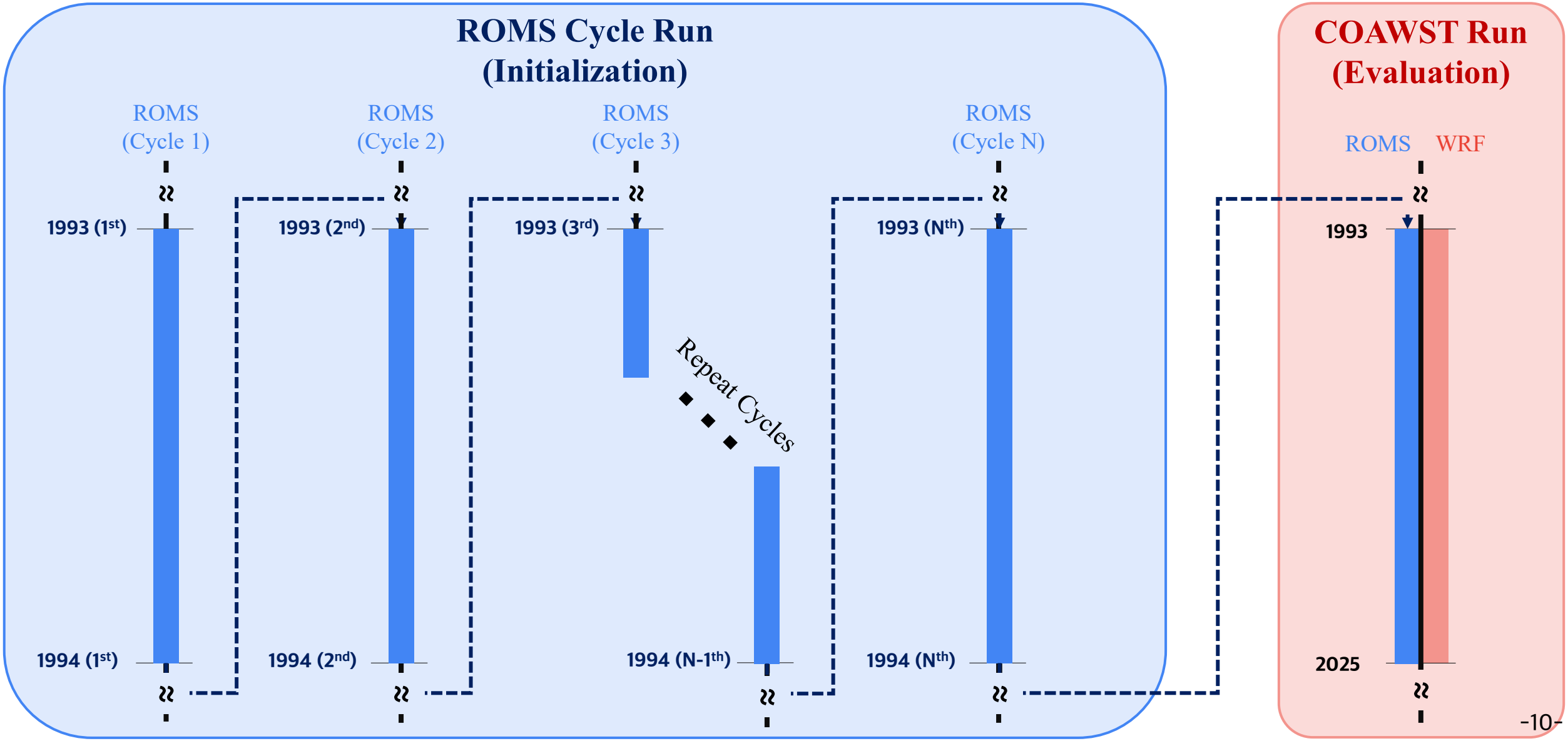
Spatial resolution: 0.1° × 0.1° (1/10°, horizontal)

Temporal extent: 1 Jan 1979 to real-time.

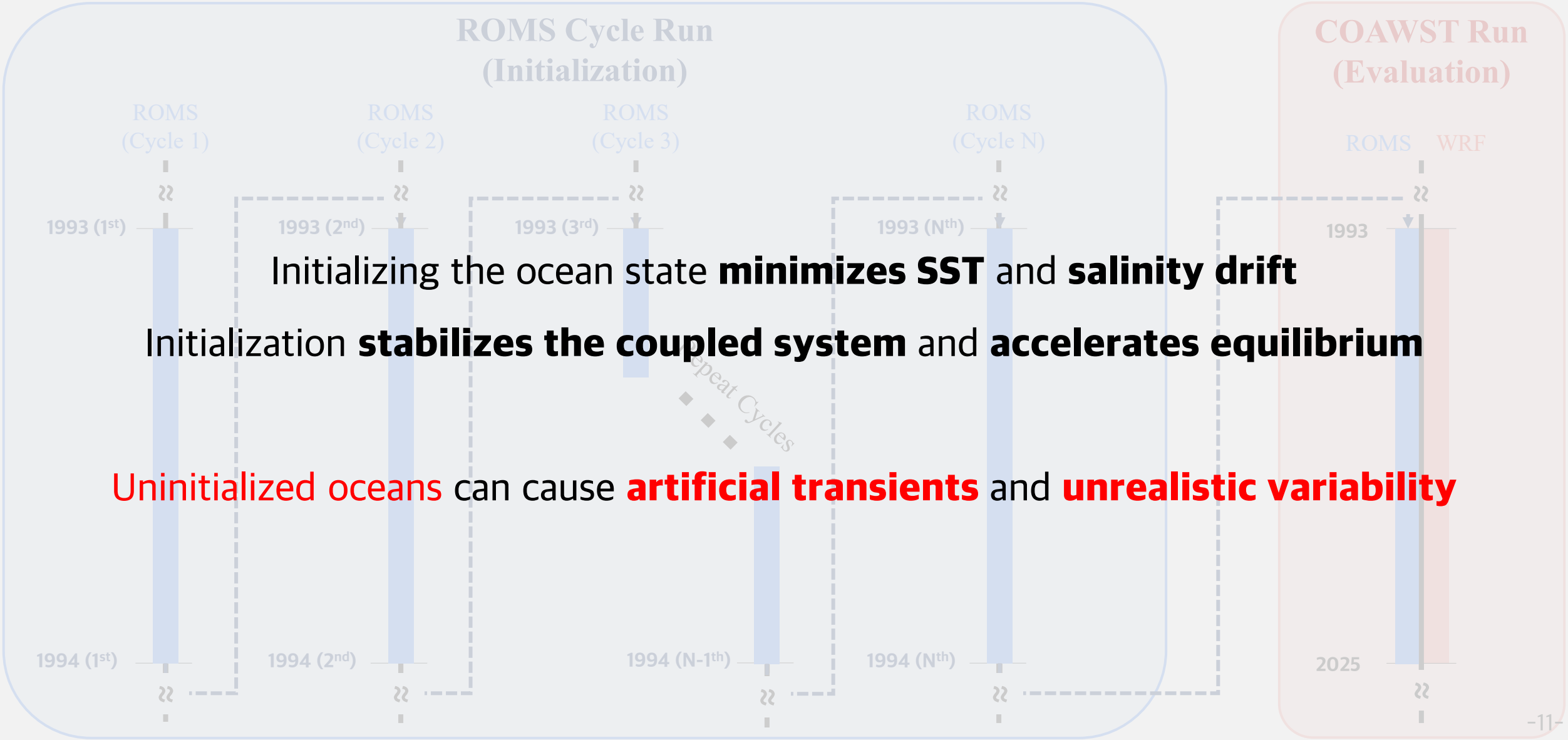
Temporal resolution: 3-hourly

*** Planning to compare the MSWEP data using GPCP (Global Precipitation Climatology Project) and validate model results.**

2-2. Ocean Initialization Method



2. Ocean Initialization Method



Initializing the ocean state **minimizes SST** and **salinity drift**

Initialization **stabilizes the coupled system** and **accelerates equilibrium**

Uninitialized oceans can cause **artificial transients** and **unrealistic variability**

2-4. Model Configuration

Couple System	Coupled Ocean-Atmosphere-Wave-Sediment Transport (COAWST)
Coupler	Model Coupling Toolkit (MCT) v 2.6.0
Ocean	Regional Ocean Modeling System (ROMS) 3.9
Atmosphere	Weather Research and Forecasting Model (WRF) v 4.2.2
Wave(s)	Simulating Waves Nearshore (SWAN) v 41.31 WAVEWATCH III (WW3) v 7.12 Infragravity wave model (InWave) v 2.0
Sediment Transport	Community Sediment Transport Modeling Systems (CSTMS)

Atmosphere	Weather Research and Forecasting Model (WRF) v 4.2.2
Horizontal Resolution| (Domain)	12 km (825×521) / dt = 36 s
Vertical Layers	35 hybrid levels
Microphysics	WSM6
Cumulus Parameterization	MSKF
Radiation	RRTMG
Planetary Boundary Layer	YSU
Land Surface	Noah LSM
Initial & Boundary data	ERA5 (0.25° and 6 h)

Ocean	Regional Ocean Modeling System (ROMS) 3.9
Horizontal Resolution | (Domain)	12 km (817×513) / dt = 30 s
Ocean depth	ETOPO2
Vertical Layers	30
Lateral boundary	Mixed Radiation Nudging: temperature, salinity, 3D U and V momentum (baroclinic) Shchepetkin: 2D U and V momentum (barotropic) Chapman_implicit: Free-surface, every 6hours
Boundary Layer Scheme (Vertical Turbulent Mixing)	Generic Length Scale (Umlauf and Burchard, 2003)
Momentum and tracer diffusion	Harmonic
River inflow / Tide	X / O
Initial & Boundary data	GLORYS12V1 (0.083° / daily)

3. Results

MSWEP
WRF
COAWST_noinit
COAWST_init

0) Observation or Reanalysis (MSWEP, GLORYS, ERA5 etc.)

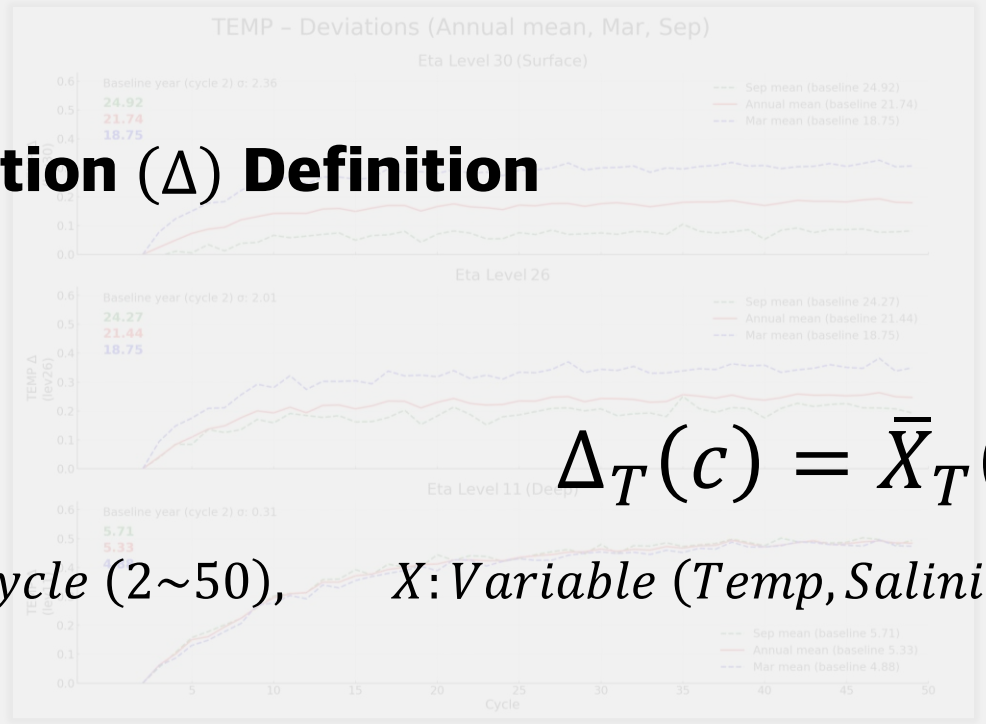
1) WRF: Uncoupled, Atmosphere-only

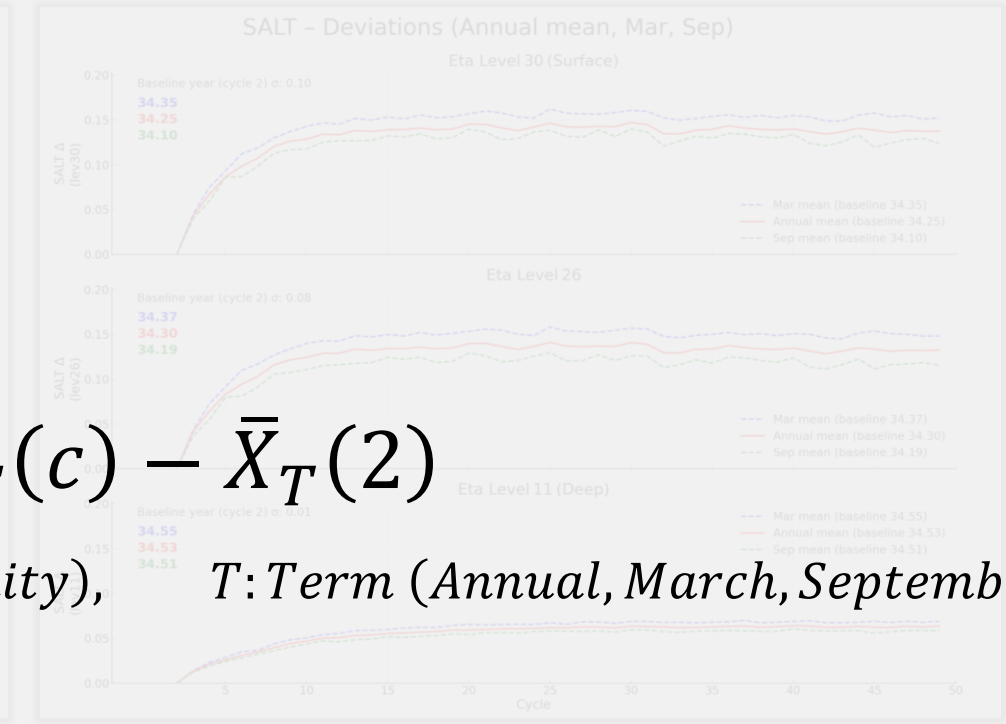
2) COAWST_noinit: Coupled, without ~~initialization~~

3) COAWST_init: Coupled, with initialization

3-1. Ocean Initialization Experiments

Deviation (Δ) Definition





$$\Delta_T(c) = \bar{X}_T(c) - \bar{X}_T(2)$$

c: $cycle$ $(2\sim50)$, X: $Variable$ $(Temp, Salinity)$, T: $Term$ $(Annual, March, September)$

[Level-wise]

- Surface & mid-depth: TEMP & SALT deviations converge ~10 cycles.
- Deep: Gradual stabilization by cycle 20–35.
- Rapid equilibrium in upper layers (<10 cycles).
- Full-depth TEMP equilibrium by ~35 cycles.
- Thresholds should be defined using observations/reanalysis.

[Seasonal]

- TEMP: Sep deviation decreases slowly → seasonal contrast reduces.
- SALT: Seasonal gap maintained → cycle run preserves seasonal signal.

Definition: Deviation from the baseline average (Cycle 2) for each term.

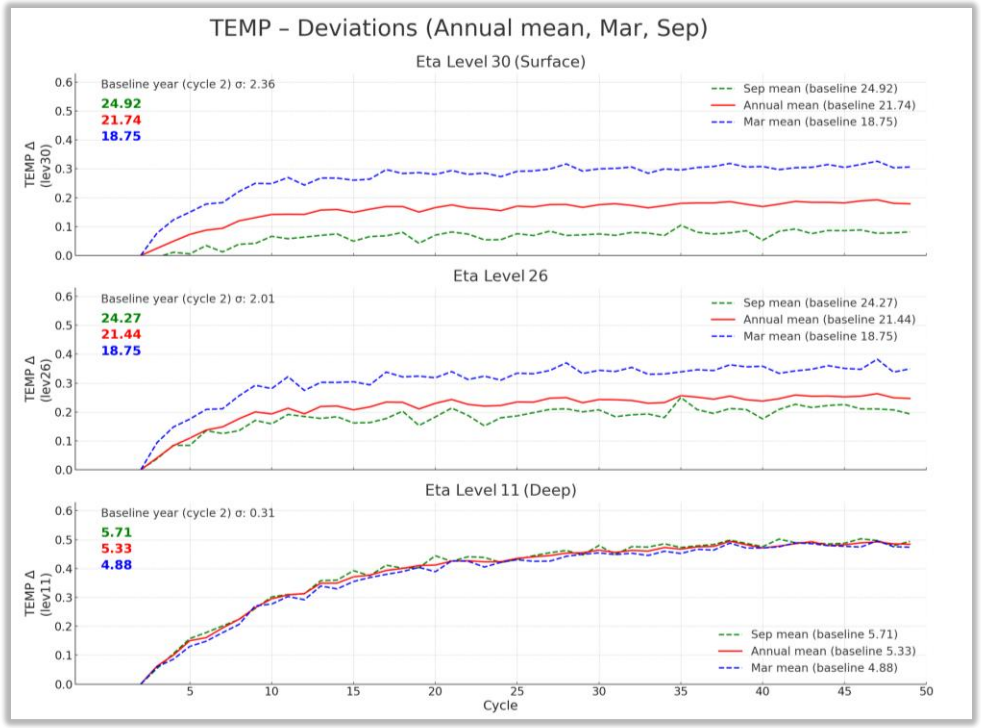
Purpose: To track convergence speed, residual deviation magnitude, and seasonal/depth-dependent characteristics by focusing on relative changes rather than absolute values.

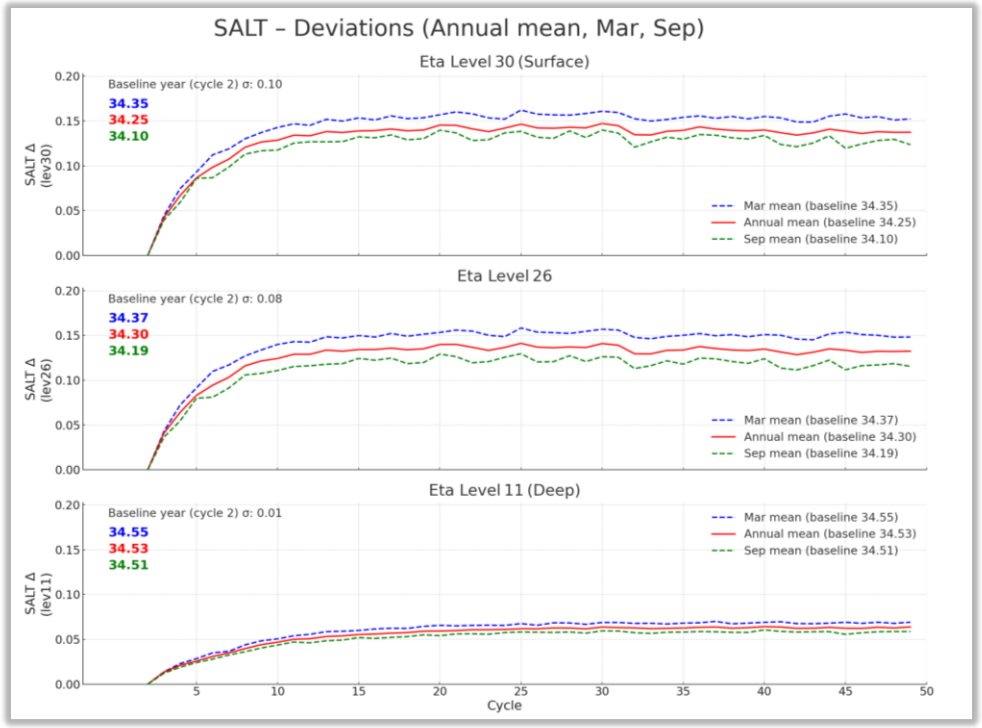
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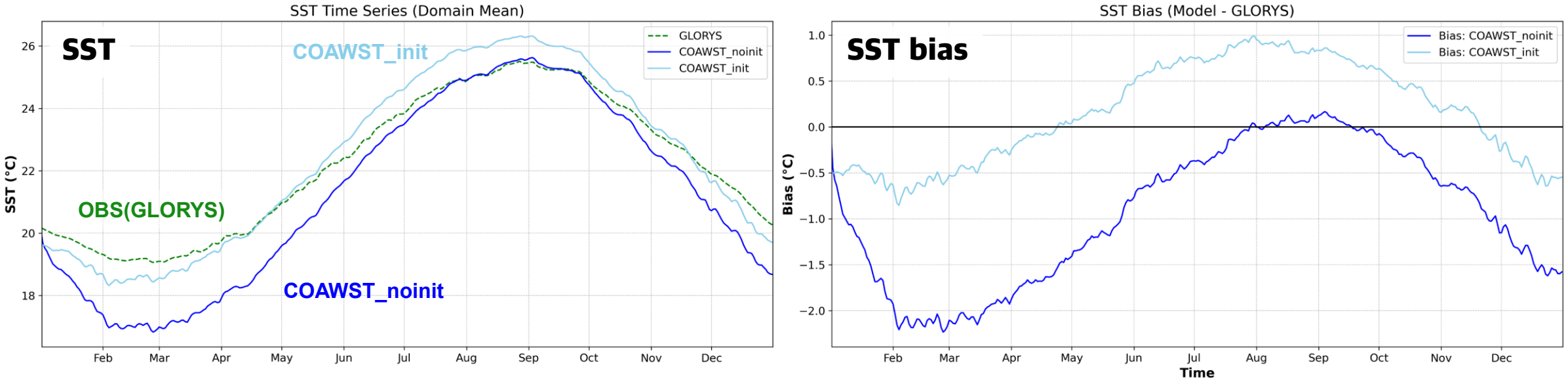
[Summary]

- Rapid equilibrium in upper layers (<10 cycles).
- Full-depth TEMP equilibrium by ~35 cycles.
- Thresholds should be defined using reanalysis statistics9op.

3-2. Sea Surface Temperature (SST)

1993 SST Time Series

Unit: ˚C



- **COAWST_init** shows much **smaller SST bias** compared to **COAWST_noinit**, especially in **winter**.
- Without initialization, **COAWST_noinit** **underestimates SST** by over 2°C early in the year.
- Initialization helps maintain **realistic SST** throughout the year, **improving model stability**.

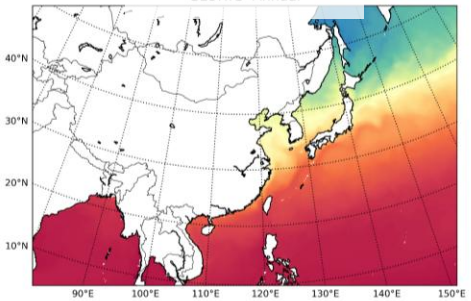
3-2. Sea Surface Temperature (SST)

1993 SST Annual Mean

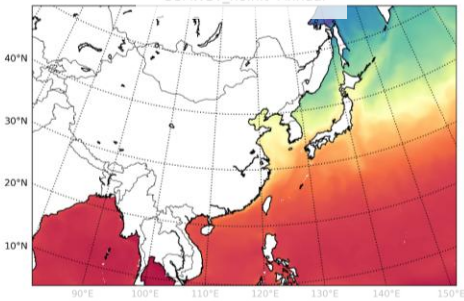
ANNUAL

Unit: ˚C

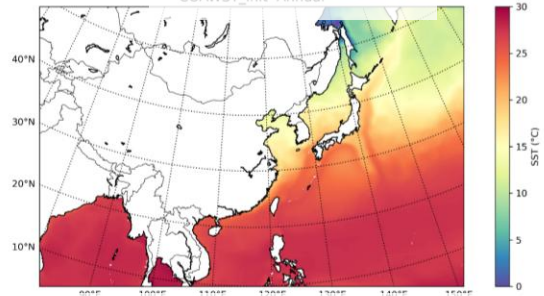
GLORYS



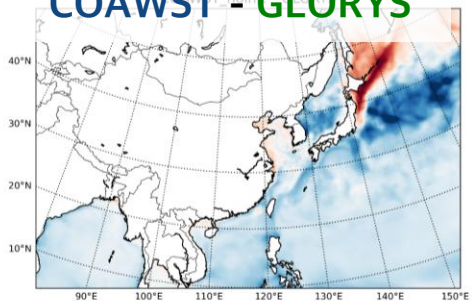
COAWST



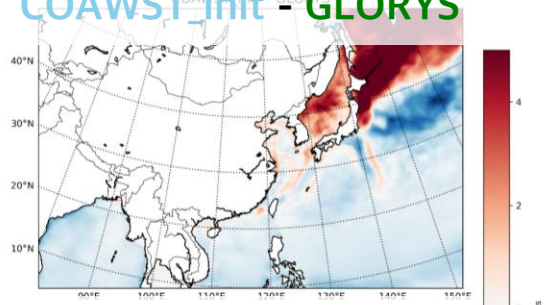
COAWST_init



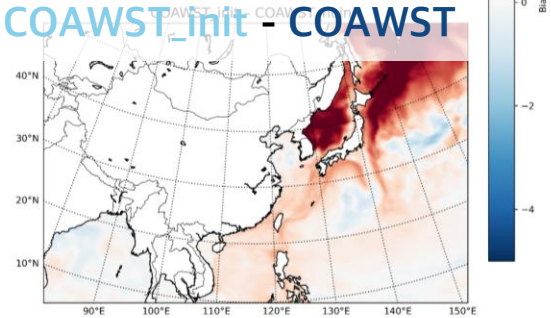
COAWST - GLORYS



COAWST_init - GLORYS



COAWST_init - COAWST



- SST is generally **underestimated**, except for a **warm bias** in the **Okhotsk Sea**.
- **COAWST_init** shows **stronger warm bias** in the Okhotsk Sea compared to **COAWST_noinit**.
- **Initialization** effectively **reduces cold bias in most other regions**.

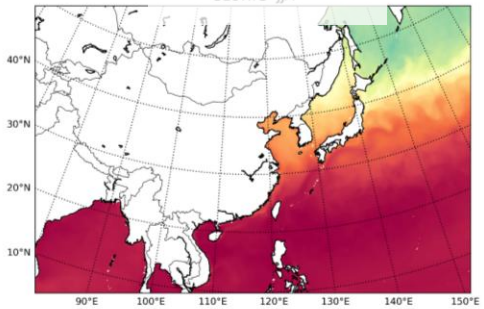
3-2. Sea Surface Temperature (SST)

1993 SST Summer (JJA) Mean

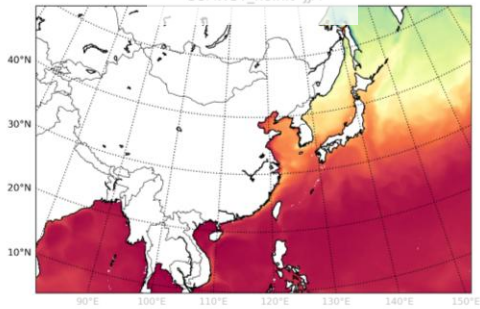
JJA

Unit: ˚C

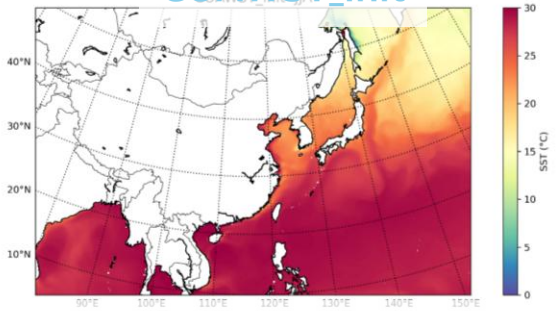
GLORYS



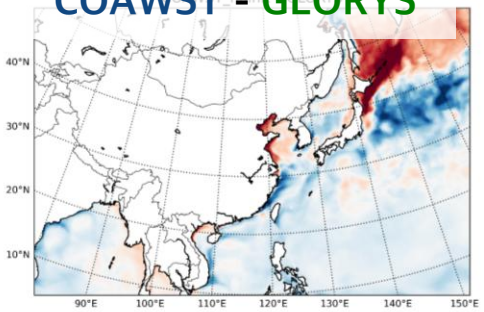
COAWST



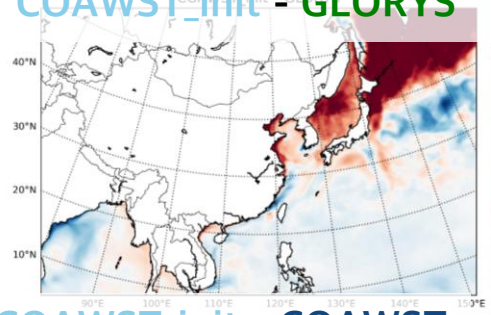
COAWST_init



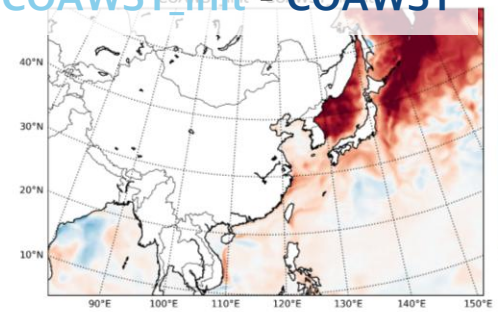
COAWST - GLORYS



COAWST_init - GLORYS



COAWST_init - COAWST

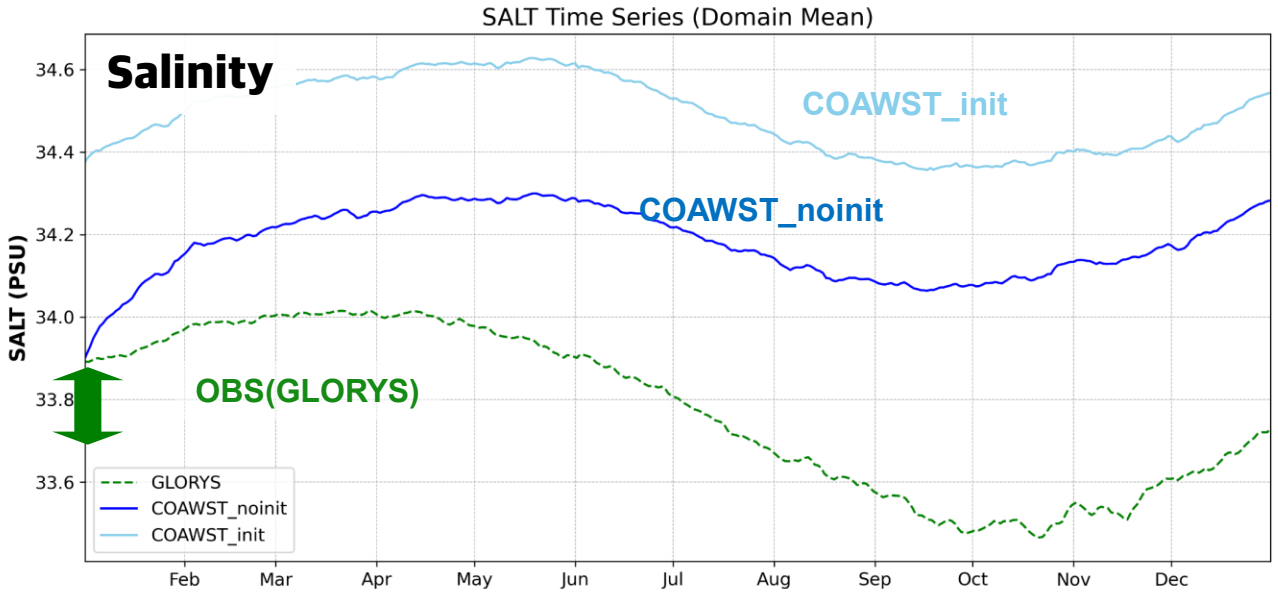


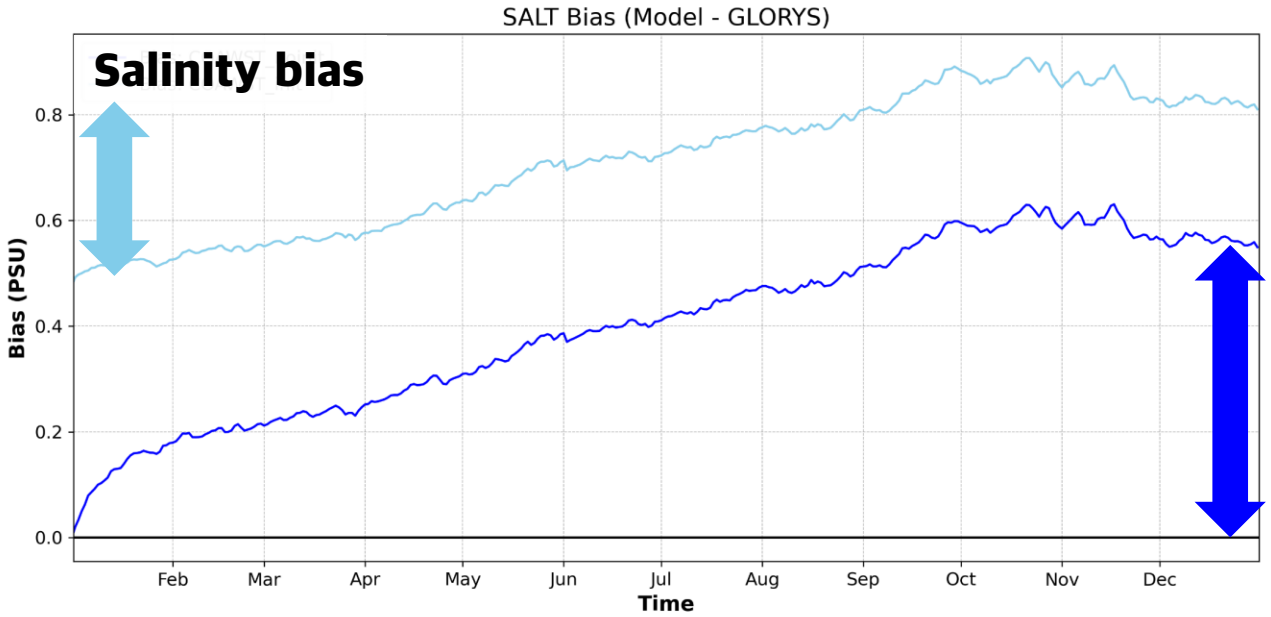
- SST bias patterns in JJA are spatially **similar to annual mean** but **more intensified**.
- **Initialization** shows **stronger impact in summer**, notably reducing cold bias.
- However, **warm bias near the Okhotsk Sea** becomes **more prominent in JJA**.

3-3. Sea Surface Salinity

1993 Salinity Time Series

Unit: PSU





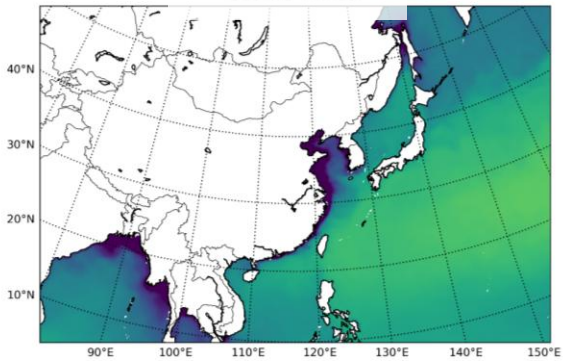
- **Salinity bias increases** gradually in **COAWST_noinit**, indicating **progressive salinization**.
- **COAWST_init** starts with **higher bias** and **maintains elevated salinity** throughout the year.
- **Seasonal variability is well captured** in both experiments despite bias levels

3-3. Sea Surface Salinity

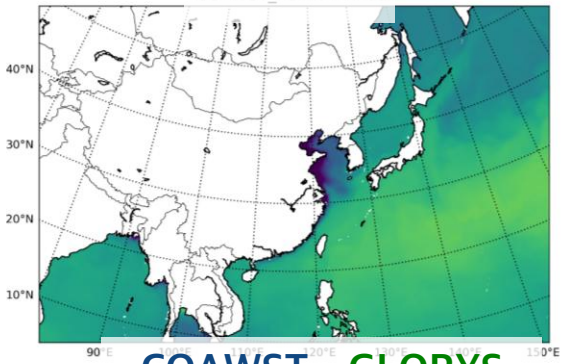
1993 Salinity Annual Mean

ANNUAL

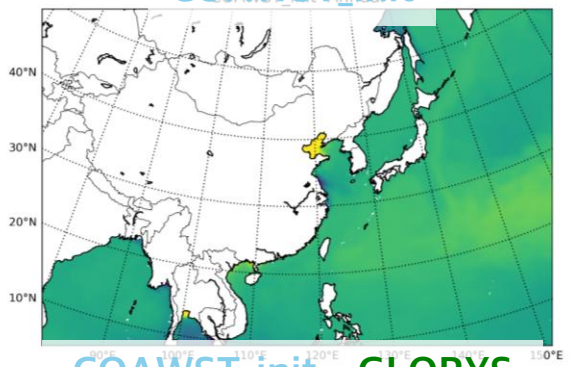
GLORYS

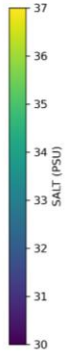


COAWST

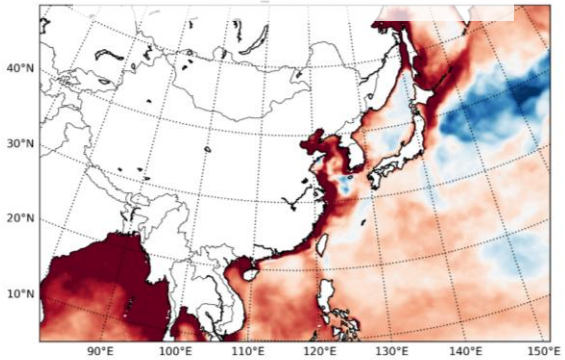


COAWST_init

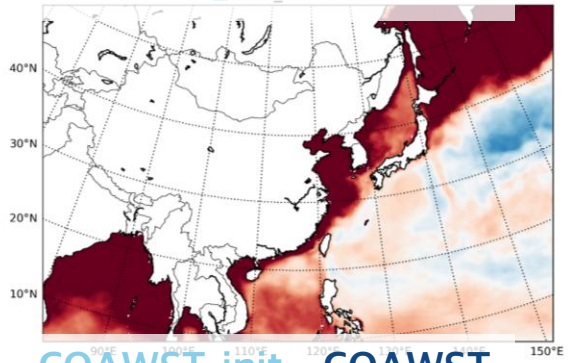




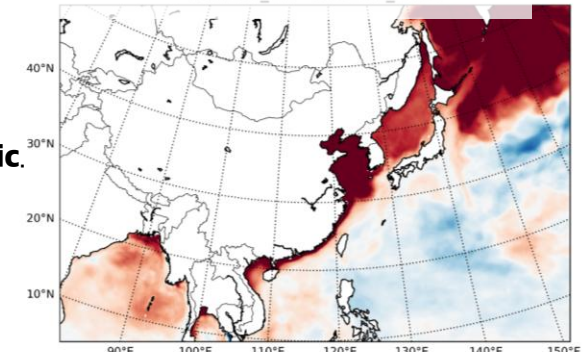
COAWST - GLORYS

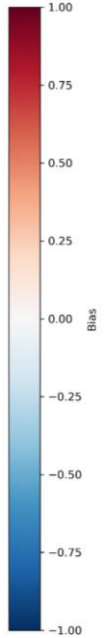


COAWST_init - GLORYS



COAWST_init - COAWST





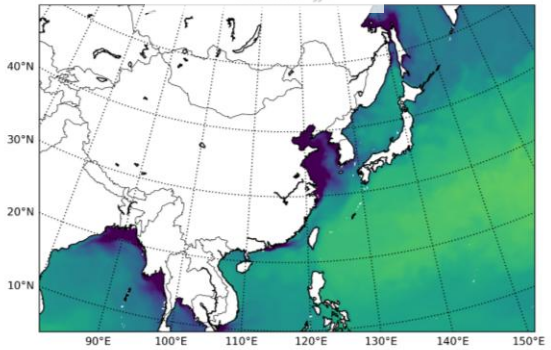
- **COAWST_noinit overestimates salinity** in most regions, except for **slight underestimation in the northwest Pacific**.
- **COAWST_init further intensifies salinity overestimation**, especially **along the coasts and in the Okhotsk Sea**.
- The impact of initialization generally leads to **saltier ocean states** across the domain.

3-3. Sea Surface Salinity

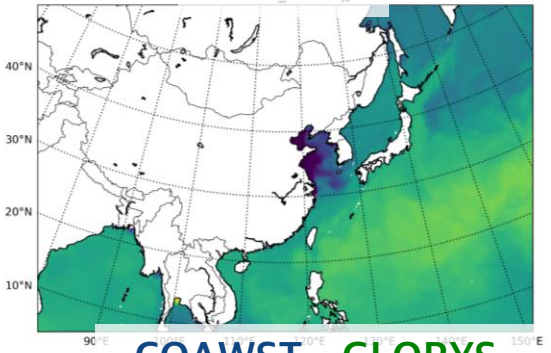
1993 Salinity Summer (JJA) Mean

JJA

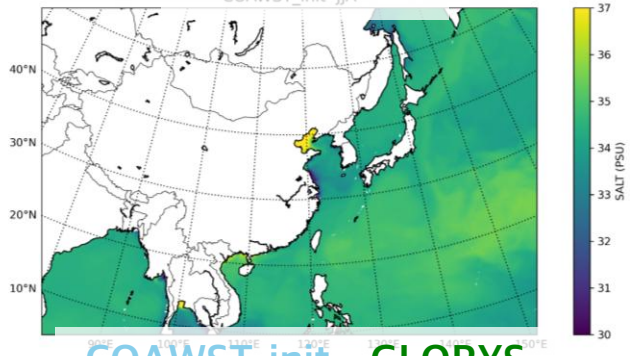
GLORYS



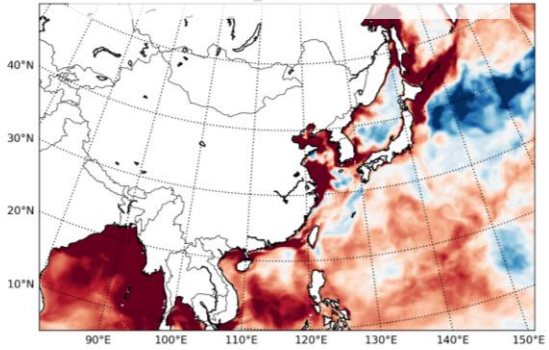
COAWST



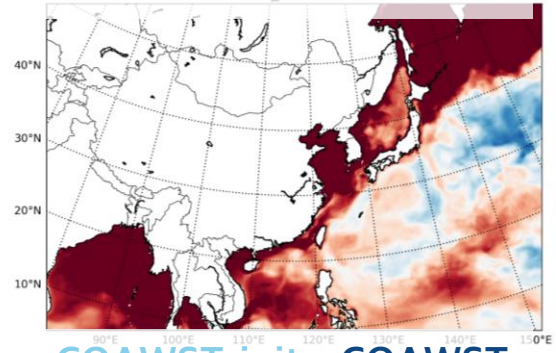
COAWST_init



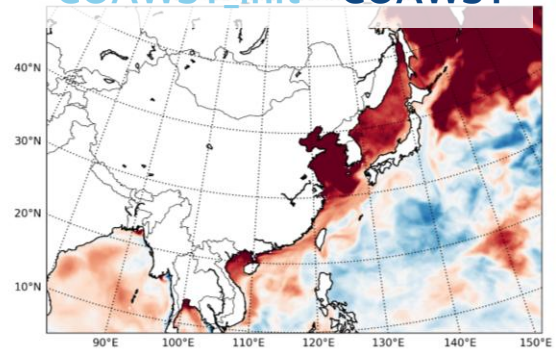
COAWST - GLORYS

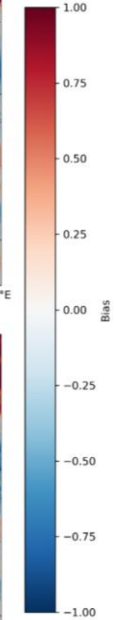


COAWST_init - GLORYS



COAWST_init - COAWST



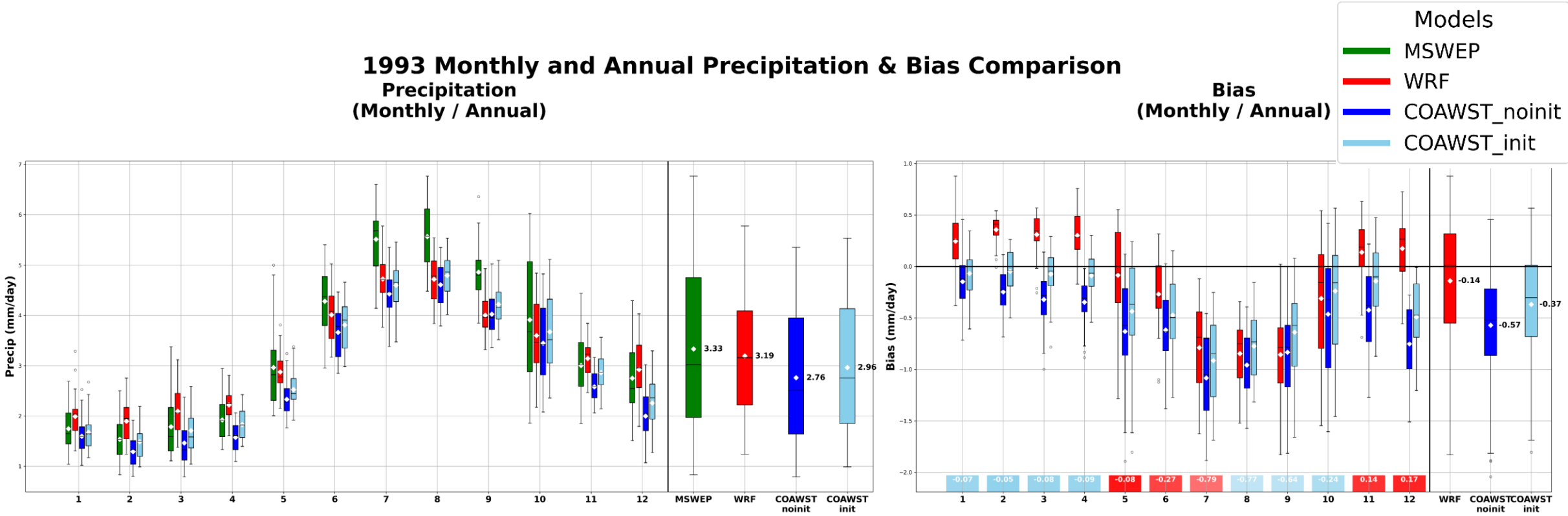


- **COAWST_noinit significantly overestimates salinity** in JJA, especially along the **western boundary** and in the **Okhotsk Sea**.
- **COAWST_init amplifies the overestimation** over **coastal regions**.
- **Salinity increases across most regions** after initialization, similar to the annual mean but **more intensified in summer**.

3-4. Precipitation Time Series Box Plot

1993 PRCP Entire Domain

Domain

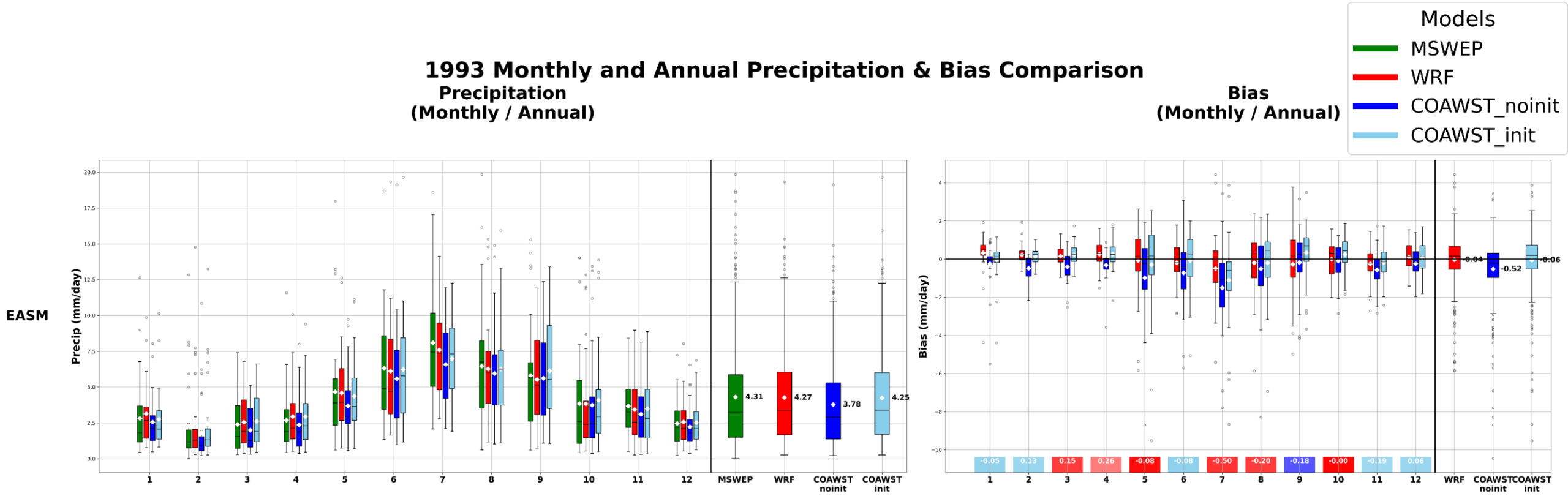


- **MSWEP**'s **seasonal variability** is generally **well captured** by the models.
- Strong or weak precipitation events are not well captured — models tend to **simulate a flattened distribution**.
- **WRF** shows the **best match** in terms of **annual mean**, but **COAWST_init** performs **better** in **monthly means**, especially in **January-April**, and **August-October**.
- **WRF** tends to **overestimate winter** and **underestimate summer-fall precipitation**.
- **COAWST_noinit** generally **underestimates** precipitation, but **COAWST_init** shows **improvement**.

- Domain: Entire area excluding the buffer zone (20 × 12 km = 240 km)
- Box plot: (Q3 + 1.5 * IQR, Q3, median, mean, Q1, Q1 - 1.5 * IQR)
- Bottom part of bias: monthly best model (lowest mean bias)

3-4. Precipitation Time Series Box Plot

1993 PRCP EASM



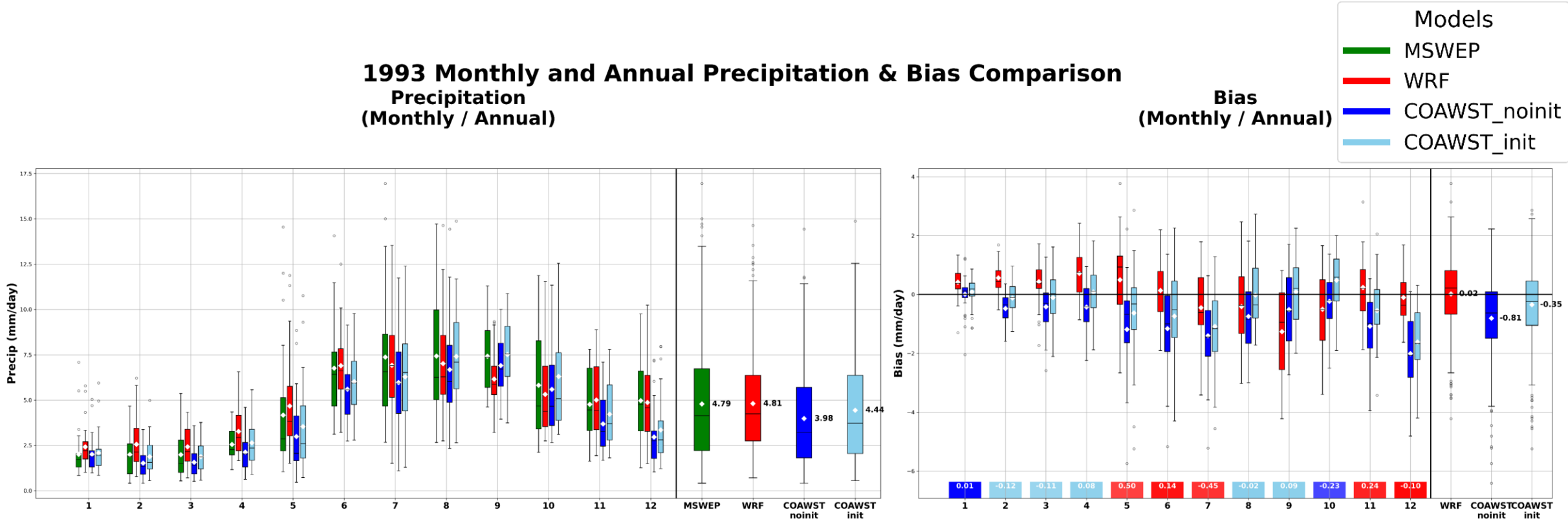
- Models **reasonably capture seasonal variability** and **strong rainfall** over the **EASM region**.
- **WRF** and **COAWST_init** show **comparable annual means**, but **COAWST_noinit underestimates** consistently.
- **Initialization** substantially **reduces precipitation bias in COAWST_noinit**, especially during **summer**

- Domain: 20°N - 40°N, 120°E - 135°
- Box plot: (Q3 + 1.5 * IQR, Q3, median, mean, Q1, Q1 - 1.5 * IQR)
- Bottom part of bias: monthly best model (lowest mean bias)

3-4. Precipitation Time Series Box Plot

1993 PRCP WNPSM

WNPSM

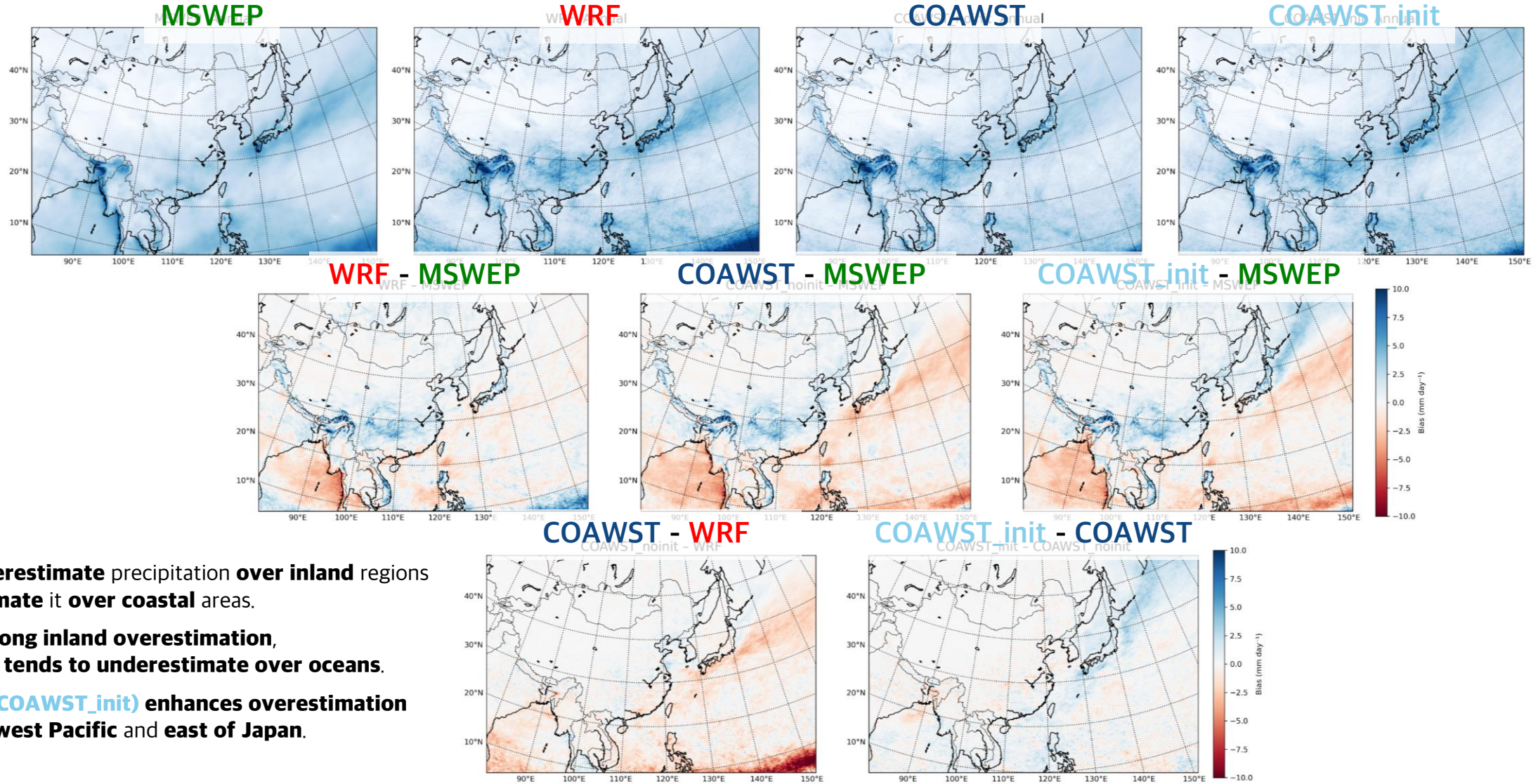


- Models capture the **seasonal pattern of WNPSM rainfall**, especially the **summer peak (JJA-SON)**.
- **COAWST_init improves precipitation** estimates significantly **over COAWST_noinit**.
- **WRF matches annual means well** but still shows **winter-spring overestimation** and **summer-fall underestimation**.

- Domain: 10°N - 30°N, 110°E - 130°
- Box plot: (Q3 + 1.5 * IQR, Q3, median, mean, Q1, Q1 - 1.5 * IQR)
- Bottom part of bias: monthly best model (lowest mean bias)

3-4. Precipitation

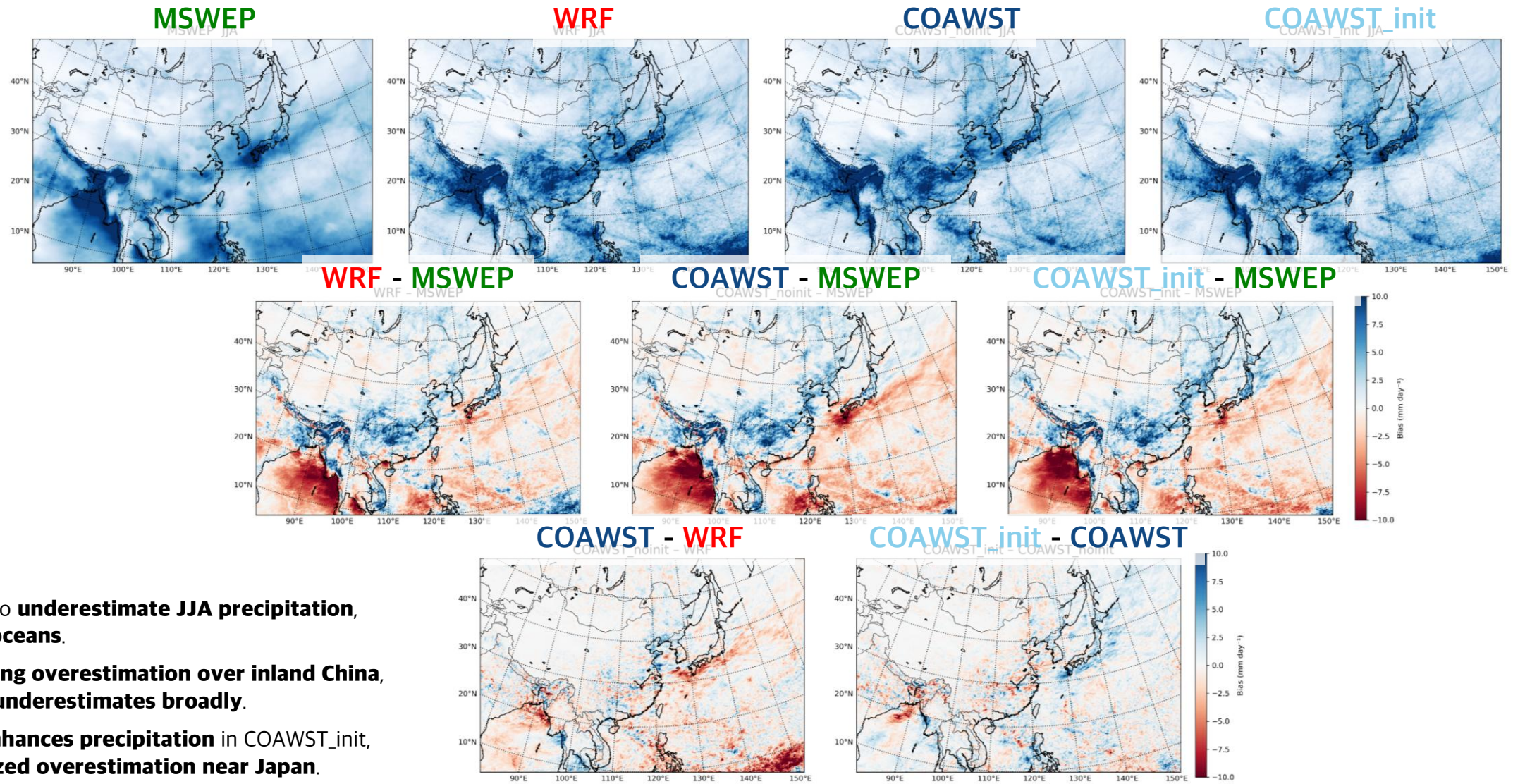
1993 PRCP Annual Mean



- **All models overestimate** precipitation **over inland** regions and **underestimate** it **over coastal** areas.
- **WRF** shows **strong inland overestimation**, while **COAWST tends to underestimate over oceans**.
- **Initialization (COAWST_init) enhances overestimation** over the **northwest Pacific** and **east of Japan**.

3-4. Precipitation

1993 PRCP Summer (JJA) Mean



- All models tend to **underestimate JJA precipitation**, especially **over oceans**.
- **WRF** shows **strong overestimation over inland China**, while **COAWST** **underestimates broadly**.
- **Initialization** **enhances precipitation** in COAWST_init, leading to **localized overestimation near Japan**.

4. Summary & Future

4-1. Summary

Ocean state matters

- Un-initialized run develops > 2 °C winter cold bias & progressive salinity drift.
- Cycle-initialized ocean sharply reduces SST/Salinity drift and accelerates equilibrium.

Coupling improves rainfall statistics but not everywhere

- Seasonal evolution & monsoon peaks reproduced; COAWST_init cuts summer bias by ~25 %.
- Winter: WRF still too wet inland; summer: all models too dry over oceans.

Remaining deficiencies

- Persistent **warm & salty bias in the Okhotsk Sea after initialization**.
- Localized coastal over-rain (Japan, East China) in COAWST_init.

Headline finding

- Air-sea interaction improves East-Asian downscaling,
  yet boundary processes (sea-ice, river runoff) and coastal dynamics now dominate the error budget.

4-2. Future

New experiment (hybrid initialization)

- **Retain the well-balanced deep-layer ocean state** from the spin-up.
- **Overwrite the top ocean layers with raw observational SST/Salinity (uncycled) to eliminate surface bias.**

Model Set-up

- Activate **interactive sea-ice in COAWST** to address Okhotsk biases.
- Add **river runoff module** for coastal fresh-water input.

Longer integrations & analysis

- Run the coupled system for multi-decadal periods to check stability and drift.
- Quantify coupling impacts on surface wind, IVT, and moisture convergence.

Additional validation

- Precipitation: bring in GPCP composites for diurnal cycle.

Thank you

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